



Design and Simulation of Network Congestion for Vehicular Networks using Streamlit & NS-3

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Aim and Objectives

Aim: To visualize network congestion in vehicular networks.

Objectives:

- Streamlit-based app: Interactive, real-time user movement simulation.
- NS-3-based visualization: Simulates realistic network behavior and logs congestion metrics such as delay, jitter, packet loss, and throughput over time.

Background:

- **Vehicular Ad Hoc Networks (VANETs):** VANETs are communication networks formed by vehicles exchanging data wirelessly. Vehicles act as mobile nodes, sharing information such as Location, Speed and some other mobility-related parameters
- **Applications of VANETs:**
 - ① **Traffic Management:** Reducing congestion and improving road safety.
 - ② **Cooperative Driving:** Enabling automated braking, collision avoidance, and better situational awareness.

Vehicle-to-Vehicle (V2V) Communication:

- V2V communication refers to the direct exchange of information between two or more vehicles on the road. This form of communication enables vehicles to share real-time data about their current status, including position, speed, acceleration, and other relevant information.

Vehicle-to-Infrastructure (V2I) Communication:

- V2I communication involves exchanging information between vehicles and roadside infrastructure components. This communication extends the capabilities of intelligent transportation systems by allowing vehicles to interact with traffic lights, road signs, and other infrastructure elements.

Importance of Visualization

- Traditional network monitoring tools may not effectively capture the unique characteristics of VANETs, such as dynamic node movement and localized congestion.
- Visualization offers:
 - Real-time insights
 - Data-driven decision-making

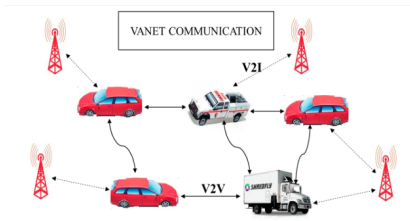
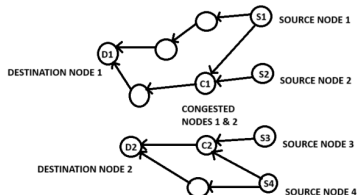


Figure: VANET Communications

- **Interactive User Interface:** Streamlit's intuitive web framework facilitates the creation of user-friendly dashboards with controls and visualizations.
- **Dynamic Network Animation:** Vehicles are represented as moving nodes on the graph, updating their positions based on simulated movement patterns.
- **User-Driven Exploration:** Users can better understand congestion patterns and their underlying causes by interacting with the controls.

Advantages:

- Intuitive and Interactive
- Real-time Visualization
- Flexibility

- **Dynamic Simulation:** Creates time-evolving models of vehicular traffic networks capturing congestion formation and dissipation patterns across 24-hour cycles.
- **Threshold-based Control:** Implements configurable congestion thresholds (50%, 70%, 80%) that trigger targeted vehicle reduction when limits would be exceeded.
- **Multi-scenario Evaluation:** Compares different parameter configurations (cap20, default, threshold50, threshold80) to quantify performance variations across strategies.

Advantages:

- Targeted Intervention During Peak Hours
- Quantifiable Parameter Optimization
- Strategic Traffic Flow Management
- Real-time Congestion Prevention

VANET Traffic Data Simulation Results

- Traffic.csv simulation provides 24-hour vehicular flow patterns with hourly granularity.

Number of Vehicles Over Time

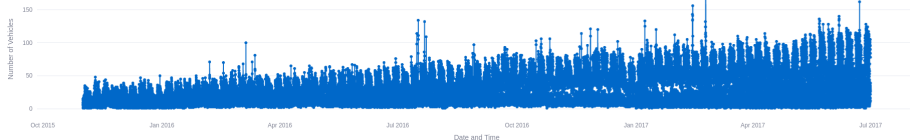


Figure: Traffic Pattern Over Time

Streamlit Approach Flow

• System Architecture and Workflow

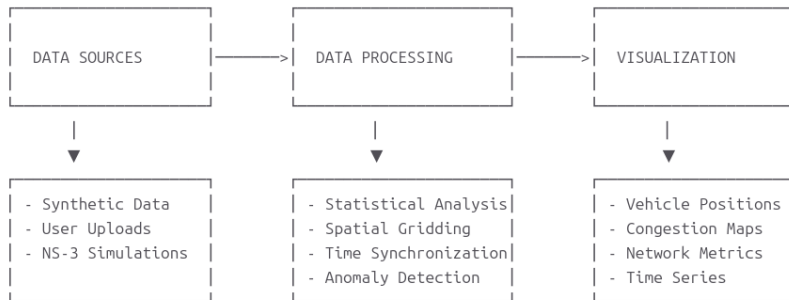


Figure: System Architecture

Streamlit Simulation Results

Data Source

This Dash_board currently have two options for working with simulation data:

- **Generate Synthetic Data :-** Create new simulation data with customizable parameters
- **Upload Data :-** Use existing NS-3 output files for visualization

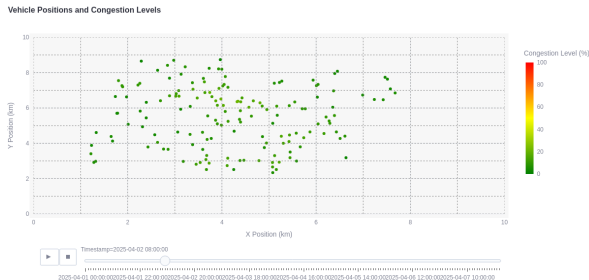


Figure: Vehicle Positions and Congestion Levels

Continue Results...

Visualizing the distribution of congestion across the network area

Congestion Heatmap

Network Congestion Heatmap

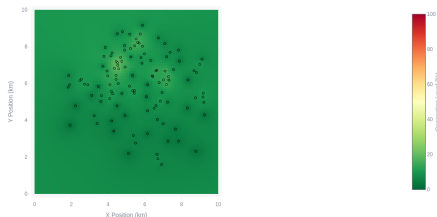


Figure: Network Congestion Heatmap

- This heatmap shows congestion intensity across different road segments. Red areas indicate high congestion, while green areas show free-flowing traffic.

Network Performance Metrics

Analysis of network flow performance indicators

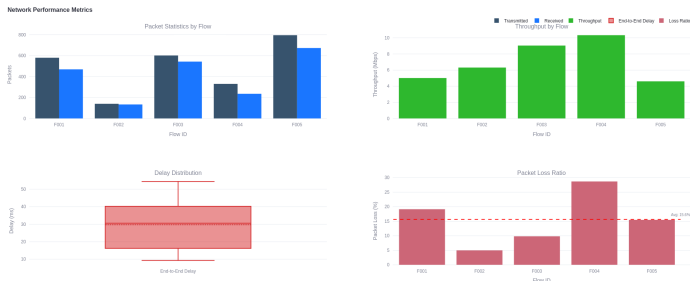


Figure: Network Performance Metrics

Time Series Analysis

- Analyzing patterns and trends in traffic data over time
- 1 For 1-Week Analysis

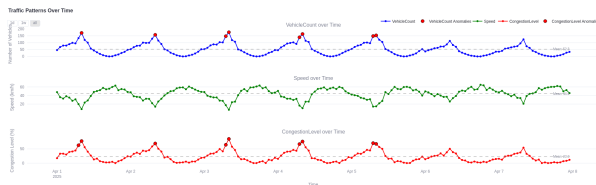


Figure: Analyzing patterns and trends in traffic data over time

Peak Congestion Analysis

- Analyzing periods of highest congestion:

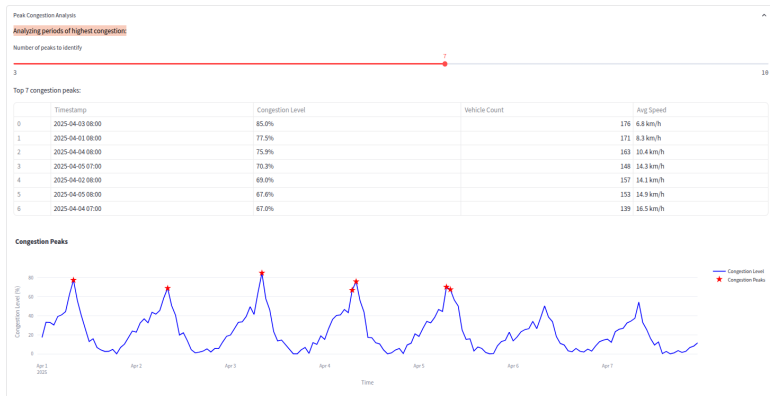


Figure: Peak Congestion Analysis

Network Signal Strength

- The network signal strength map of Bangalore visualizes signal categories—green for strong, yellow for medium, and red for weak—indicating higher signal strength near the city center and degradation towards the outskirts. Blue markers represent tower locations, showing their influence on signal distribution.

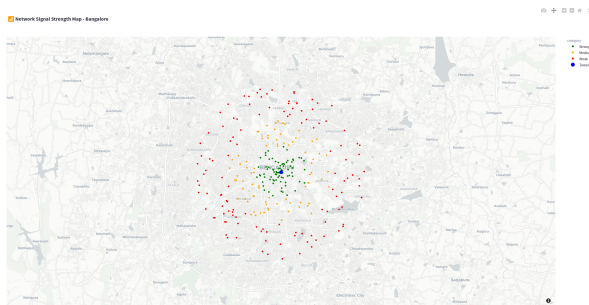


Figure: Network Signal Strength Map of Bangalore

NS-3 Approach Flow

1. NS-3 Installation

```
$ wget https://www.nsnan.org/releases/ns-allinone-3.42.tar.bz2
$ tar -xjf ns-allinone-3.42.tar.bz2
$ cd ns-allinone-3.42
$ ./build.py --enable-examples --enable-tests
```

Figure: NS-3 installation

2. Project Setup

```
$ cd ns-3.42
$ mkdir -p scratch/traffic-congestion/data
$ mkdir -p scratch/traffic-congestion/results
$ mkdir -p scratch/traffic-congestion/plots
```

Figure: Setup

3. Data Preparation:

- Place traffic CSV data file in:
scratch/traffic-congestion/data/
- Required CSV format: DateTime, Junction, Vehicles, ID

4. Congestion Control Algorithm

```
int ApplyCongestionControl(int vehicles, double congestionThreshold, int capacity) {
    double congestion = CalculateCongestion(vehicles, capacity);

    if (congestion <= congestionThreshold) {
        // No need to apply control
        return vehicles;
    }

    // Calculate max allowed vehicles based on threshold
    int maxAllowed = static_cast<int>(capacity * congestionThreshold / 100.0);

    // Apply control: limit to max allowed
    return maxAllowed;
}
```

NS-3 Simulation Result

- This NS-3 simulation analyzes traffic congestion using a dataset (traffic.csv, comparing original vs. controlled vehicle counts hourly. The simulation calculates average congestion before and after control, showing a slight reduction of 1.33%. Results, including vehicle statistics and congestion rates, are saved to simple_results.csv.

```
mritunjay@mritunjay: /ns-3.42/ns-allinone-3.42/ns-3.42$ ./ns3 run "simple-traffic --datafile=scratch/traffic-congestion/data/traffic.csv --outputfile=scratch/traffic-congestion/results/simple_results.csv --threshold=70 --capacity=25"
[9/2] We checking globbed directories...
ninja: no work to do.
Starting traffic simulation
Loading data from scratch/traffic-congestion/data/traffic.csv
Loaded 48120 data points
Simulating hour 0: 2015-11-01 00:00:00 - Original: 15 vehicles, Controlled: 15 vehicles
Simulating hour 1: 2015-11-01 01:00:00 - Original: 13 vehicles, Controlled: 13 vehicles
Simulating hour 2: 2015-11-01 02:00:00 - Original: 10 vehicles, Controlled: 10 vehicles
Simulating hour 3: 2015-11-01 03:00:00 - Original: 7 vehicles, Controlled: 7 vehicles
Simulating hour 4: 2015-11-01 04:00:00 - Original: 9 vehicles, Controlled: 9 vehicles
Simulating hour 5: 2015-11-01 05:00:00 - Original: 6 vehicles, Controlled: 6 vehicles
Simulating hour 6: 2015-11-01 06:00:00 - Original: 9 vehicles, Controlled: 9 vehicles
Simulating hour 7: 2015-11-01 07:00:00 - Original: 8 vehicles, Controlled: 8 vehicles
Simulating hour 8: 2015-11-01 08:00:00 - Original: 11 vehicles, Controlled: 11 vehicles
Simulating hour 9: 2015-11-01 09:00:00 - Original: 12 vehicles, Controlled: 12 vehicles
Simulating hour 10: 2015-11-01 10:00:00 - Original: 15 vehicles, Controlled: 15 vehicles
Simulating hour 11: 2015-11-01 11:00:00 - Original: 17 vehicles, Controlled: 17 vehicles
Simulating hour 12: 2015-11-01 12:00:00 - Original: 16 vehicles, Controlled: 16 vehicles
Simulating hour 13: 2015-11-01 13:00:00 - Original: 15 vehicles, Controlled: 15 vehicles
Simulating hour 14: 2015-11-01 14:00:00 - Original: 16 vehicles, Controlled: 16 vehicles
Simulating hour 15: 2015-11-01 15:00:00 - Original: 12 vehicles, Controlled: 12 vehicles
Simulating hour 16: 2015-11-01 16:00:00 - Original: 12 vehicles, Controlled: 12 vehicles
Simulating hour 17: 2015-11-01 17:00:00 - Original: 16 vehicles, Controlled: 16 vehicles
Simulating hour 18: 2015-11-01 18:00:00 - Original: 17 vehicles, Controlled: 17 vehicles
Simulating hour 19: 2015-11-01 19:00:00 - Original: 20 vehicles, Controlled: 17 vehicles
Simulating hour 20: 2015-11-01 20:00:00 - Original: 17 vehicles, Controlled: 17 vehicles
Simulating hour 21: 2015-11-01 21:00:00 - Original: 19 vehicles, Controlled: 17 vehicles
Simulating hour 22: 2015-11-01 22:00:00 - Original: 20 vehicles, Controlled: 17 vehicles
Simulating hour 23: 2015-11-01 23:00:00 - Original: 15 vehicles, Controlled: 15 vehicles
Running simulation...
Simulation complete!
Average original congestion: 54.5%
Average controlled congestion: 53.1667%
Congestion reduction: 1.3333%
Original vehicles: 327, Controlled vehicles: 319
Vehicles reduced: 8
Results saved to: scratch/traffic-congestion/results/simple_results.csv
mritunjay@mritunjay: /ns-3.42/ns-allinone-3.42/ns-3.42$
```

Figure: Enter Caption

Analyzing Results

- Traffic Congestion Trends Analysis When reviewing the "Traffic Congestion: Original vs. Controlled" graph:

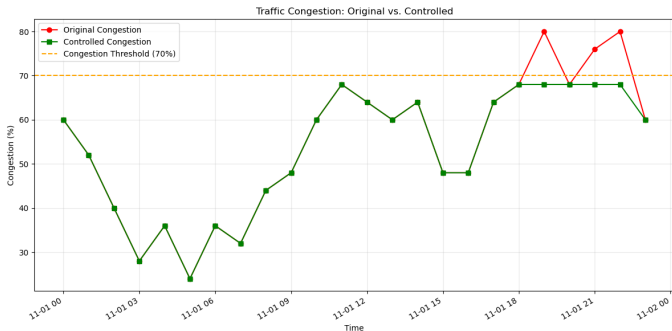


Figure: Traffic Congestions

Threshold Management Analysis

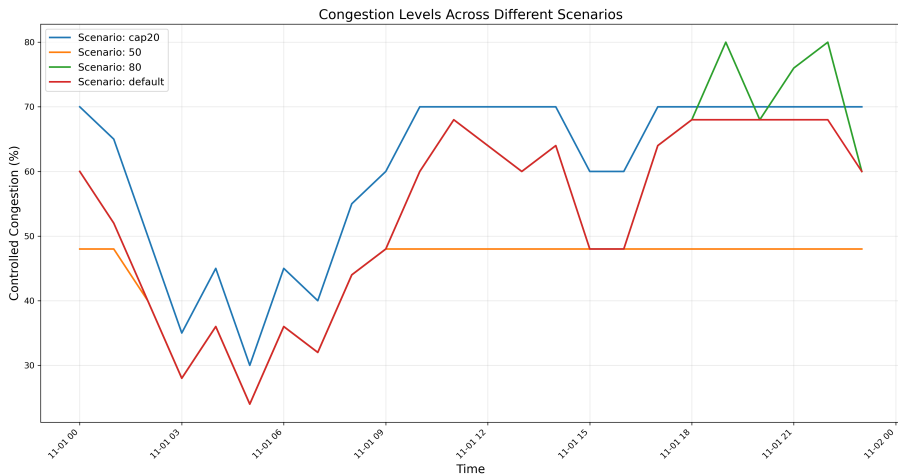


Figure: Traffic Congestion Management

Continue...

- The plots compare traffic control strategies (cap20, 50, 80, default), showing that the 'cap=50' configuration achieves the highest congestion reduction (10.5%) and most vehicle reduction (63). In contrast, 'cap=80' yields no improvement, while the default setting offers minimal benefits.

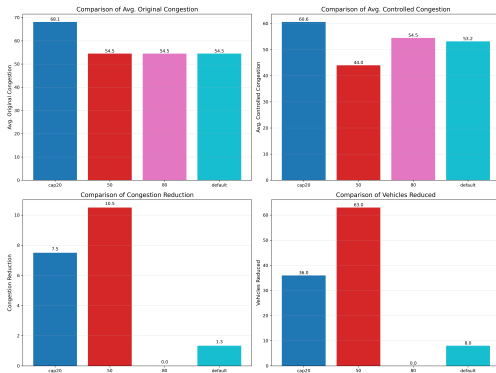


Figure: Matrics Comparison

Result and Conclusion

Both methods enable comprehensive vehicular congestion simulation through sophisticated node generation and multi-dimensional graph plotting.

- NS-3 simulation identified 70% threshold as optimal, requiring minimal intervention (8 vehicles) while preventing network congestion.
- Evening hours (18:00-22:00) demonstrated highest congestion sensitivity, while 21/24 hours needed no intervention.
- Lower thresholds (50%) achieved maximum congestion reduction (10.5%) but required significantly more vehicle reduction (63).
- The NS-3/Streamlit integration provides both powerful simulation capabilities and intuitive visualization for effective traffic management policy development.

References I

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THANK YOU!
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